

One-Pager: Problem & Solution

WHO HURTS MOST

Students across Africa struggling with textbooks and generic AI explanations.

The pain: Textbooks are 400-800 pages. Students are intimidated and don't read them. Instead, they photograph specific pages and ask generic AI assistants (ChatGPT, Claude, etc.) to explain. These assistants give one-size-fits-all answers that conflict with what their professor taught and what their textbook actually emphasizes. Students are more confused after asking AI than before.

Real scenario: A FUTO student asks a generic AI about photosynthesis. Gets an explanation. But their Biology textbook emphasizes the Calvin cycle differently, and their professor highlighted different equations during lectures. The answer is technically correct but completely useless for their actual exam because it doesn't match their curriculum.

When it hurts: Every day. Every assignment. Every exam prep.

Who it hurts: Every student who relies on textbooks and generic AI (billions across Africa).

Why it matters: Students fail exams despite studying hard because they don't actually understand — they just heard explanations, but in ways their brain doesn't click with.

THE CORE PROBLEMS

Problem 1: Textbooks Are Too Big, So Students Don't Read Them

Textbooks are 400-800 pages. Students open them, get overwhelmed by size and density, and give up. Instead of searching their textbook, they take photos of pages and ask generic AI. But generic AI doesn't know their specific curriculum, so answers conflict with textbook content and professor emphasis.

Problem 2: Explanations Aren't Personalized

Some students learn through analogies. Others need step-by-step logic. Others need visual diagrams. Generic AI explains everything the same way to every student. Half the class understands, half doesn't — even though they're asking the same question about the same topic.

Problem 3: Generic AI Hallucinates About Curriculum

Generic AI confidently explains things wrong. It doesn't know what YOUR textbook says. It doesn't know what your professor emphasized. Students get answers that conflict with their notes and textbook. They don't know who to trust anymore.

Problem 4: No One Tests If Learning Actually Happened

A student gets an explanation and moves on. They fail the exam. Why? Because nobody tested them. Nobody identified that they understood the WHAT but not the WHY. Every conversation resets. It's explanation theater, not actual learning.

THE SOLUTION

An AI tutor that:

- Retrieves explanations from YOUR textbook (not generic AI)
- Explains concepts in the way YOUR brain works (learns your learning style)
- Tests you to verify you actually understand (not just heard it)
- Remembers what confused you before (builds your persistent learning profile)
- Improves every interaction (gets smarter as you use it)

Result: Every student gets explanations tailored to how THEY learn, grounded in THEIR curriculum, with proof they actually learned it.

VALIDATION APPROACH

We will conduct 5+ conversations with real FUTO students to validate:

1. "Does generic AI frustrate you when you study?"
2. "Would you use a tutor that adapted to how you learn?"
3. "Would you trust explanations from your actual textbook more than generic AI?"
4. "If this system helped you understand better, would you use it instead of asking friends or generic AI?"
5. "What's your biggest frustration with studying right now?"

Expected result: 5/5 students confirm the problem is real. At least 4/5 say they'd use this system.